$$\frac{AE}{L}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}\begin{Bmatrix}u_1\\ u_2\end{Bmatrix}=\begin{bmatrix}\frac{Lq_0}{2}\\ \frac{Lq_0}{2}\end{bmatrix}+\begin{bmatrix}R\\ F_1^1\end{bmatrix}\text{Element 1}$$

$$\frac{AE}{L}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}\begin{Bmatrix}u_2\\ u_3\end{Bmatrix}=\begin{bmatrix}\frac{Lq_0}{2}\\ \frac{Lq_0}{2}\end{bmatrix}+\begin{bmatrix}F_1^2\\ F_2^1\end{bmatrix}\text{Element 2}$$

$$\frac{AE}{L}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}\begin{Bmatrix}u_3\\ u_4\end{Bmatrix}=\begin{bmatrix}F_2^2\\ F_3^1\end{bmatrix}\text{Element 3}$$

$$\frac{AE}{L}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}\begin{Bmatrix}u_4\\ u_5\end{Bmatrix}=\begin{bmatrix}F_3^2\\ F_4^1\end{bmatrix}\text{Element 4}$$

$$F_1^2+F_1^1=f_1$$

$$\frac{AE}{L}\begin{bmatrix}1 & -1 & 0 & 0 & 0\\ -1 & 2 & -1 & 0 & 0\\ 0 & -1 & 2 & -1 & 0\\ 0 & 0 & -1 & 2 & -1\\ 0 & 0 & 0 & -1 & 1\end{bmatrix}\begin{Bmatrix}u_1\\ u_2\\ u_3\\ u_4\\ u_5\end{Bmatrix}=\begin{bmatrix}125\\ 250\\ 125\\ 0\\ 0\end{bmatrix}+\begin{bmatrix}R\\ 0\\ 0\\ 500\\ 1000\end{bmatrix}$$

$$\frac{AE}{L}\begin{bmatrix}1 & -1 & 0 & 0 & 0\\ -1 & 2 & -1 & 0 & 0\\ 0 & -1 & 2 & -1 & 0\\ 0 & 0 & -1 & 2 & -1\\ 0 & 0 & 0 & -1 & 1\end{bmatrix}\begin{Bmatrix}u_1\\ u_2\\ u_3\\ u_4\\ u_5\end{Bmatrix}=\begin{bmatrix}125\\ 250\\ 125\\ 0\\ 0\end{bmatrix}+\begin{bmatrix}R\\ 0\\ 0\\ 500\\ 1000\end{bmatrix}$$

$$\frac{0.01*2*10^{11}}{0.25}\begin{bmatrix}2 & -1 & 0 & 0\\ -1 & 2 & -1 & 0\\ 0 & -1 & 2 & -1\\ 0 & 0 & -1 & 1\end{bmatrix}\begin{Bmatrix}u_2\\ u_3\\ u_4\\ u_5\end{Bmatrix}=\begin{bmatrix}250\\ 125\\ 0\\ 0\end{bmatrix}+\begin{bmatrix}0\\ 0\\ 500\\ 1000\end{bmatrix}$$

$$\begin{Bmatrix}u_2\\ u_3\\ u_4\\ u_5\end{Bmatrix}=\begin{bmatrix}2.34375E-7\\ 4.375E-7\\ 6.25E-7\\ 7.5E-7\end{bmatrix}$$

$$Stress\!:\sigma=\frac{F}{A};u_x=\frac{F}{K};K=\frac{AE}{L}$$

$$K=8E9;A=0.01;F=u_x*K;\ Stress=\sigma=\frac{u_x*8E9}{0.01}=u_x*8E11$$

These are NOT max Stresses. These are the stresses at the center point of each element.

$$\begin{Bmatrix}\sigma_1\\ \sigma_2\\ \sigma_3\\ \sigma_4\end{Bmatrix}=\begin{bmatrix}2.34375E-7*8E11\\ (4.375E-7-2.34375E-7)*8E11\\ (6.25E-7-4.375E-7)*8E11\\ (7.5E-7-6.25E-7)*8E11\end{bmatrix}=\begin{bmatrix}2.34375*8E4\\ 2.03125*8E4\\ 1.875*8E4\\ 1.25*8E4\end{bmatrix}=\begin{bmatrix}187.5\\ 162.5\\ 150\\ 100\end{bmatrix}kPa$$